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055697 – Numerical Analysis For Machine Learning
Project Report

Facial Expression Recognition

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1 Abstract

This project replicates and extends a published facial expression recognition pipeline using the Facial Expression Recognition (FER) dataset published on Kaggle [1] composed of 35,887 facial images across seven emotions. Following the proposed methodology, the dataset is cleaned by identifying and removing blank/noisy samples, with additional strategies to get a cleaner dataset. The resulted dataset is normalized and split into training, validation and testing subsets. Using these data as a base, three models are implemented and compared: (1) a multi-class SVM (Support Vector Machine) classifier based on a PCA feature representation of the images, (2) a convolutional neural network based on a VGG19 architecture, and (iii) a sequential Convolutional Neural Network variant motivated by the framework of hypothesis-testing of the paper, to reduce confounding errors between visually similar expressions.

The performance of the models was evaluated using accuracy, precision, recall, and F1-score, complemented by confusion matrices. These metrics provide a safeguard against misleading performance that can arise from naive strategies—such as predicting majority classes or exploiting class imbalance—by checking whether each emotion class is learned with consistent precision/recall rather than being guessed opportunistically.

These implementation results indicate that a CNN based on VGG19 architecture provides the highest predictive performance, aligning with the paper's conclusion that convolutional architectures are well-suited to the hierarchical visual features present in facial expression images. This supports the broader claim that the selected Dataset contains learnable spatial patterns that a CNN can leverage to separate emotion classes more effectively than classical feature reduction and shallow classifiers.

2 Introduction

In contemporary machine learning, two areas of application that have attracted substantial attention in recent years are the face recognition task and the sentiment analysis problem. Facial expression recognition sits at the intersection of these areas. It is a type of biological method that can accurately analyze and evaluate facial patterns in facial contours to uniquely identify and verify a person. Face recognition has become a widely used and widely recognized biometric and has grown significantly in security-related areas. In terms of the least expensive and most commonly used touch-less biometrics, it has gained more importance during pandemic situations [3].

Despite its relevance, this recognition task remains a challenging problem. Real-world facial images exhibit substantial variability in pose, illumination, quality, and subject-specific differences (expressions of the face and individual variations in facial structure.). Not to mention that the boundaries between emotion categories are itself ambiguous, since each expression is different for each person, and the interpretation of them is also depen-

dent. This leads to a inter-class overlap and increases the likelihood of noise in the labeled dataset and, therefore, in the predictions of the models.

This project aims to replicate and refine the methodology and models proposed in the paper "Facial Image expression recognition and prediction system", to verify and discuss the final thoughts on the dataset studied. Following the reference pipeline, first dataset preparation and cleaning is performed, including an additional strategy to identify and remove samples that are unlikely to provide useful information during training. Then, the 3 models suggested are implemented, starting with a lightweight model based on an SVM using just the first 70 Principal Components of each image. Then, from scratch, the VGG19-like model is implemented to accomplish the multiclass classification, and finally, a simpler convolutional model is implemented to examine the stated hypothesis regarding model behavior.

3 Problem statement and paper summary

The paper [3] proposes a facial expression recognition framework based on the following methodology.

- **Data collection and preprocessing**

  The data source used consists of various photographs captured and published on Google and different online platforms, with any format and any color scale. This part starts by analyzing the dataset, removing blank images, and performing a dimensional reduction using Principal Component Analysis. At the same time, a train-test split is performed to format the data and be prepared to feed the models.

  In this stage, a strategy to remove "blank images" and any kind of images that could be categorized as noise is included, since the paper does not specify the complete criteria to remove these images. This new strategy is based on the computation of any valuable metric over the pixels values and filtering the data based on a threshold. This strategy and a manual inspection of the filtered data allows to remove almost all the not valuable data before starting the model training stage.

- **Model Design and selection**

  There are 3 models proposed based on the preprocessed data and any reduction performed. The first model is an SVM trained on PCA-reduced image representations. The second and third models leverage Convolutional Neural Networks (CNNs). **Model-2** follows a VGG19 architecture with and additional that outputs predictions across the seven classes (Emotions). The third model uses a convolutional sequential layer to perform a testing hypothesis about the performance and validity of the previous models in this classification task with the data supplied.

In this stage, each model was implemented from scratch and trained with the specifications mentioned in the paper, and tuned for the parameters not mentioned. Therefore, the results are not exactly the same, and it was searched to improve any aspect of the model training, but the behavior follows the same trend as the one published in the paper.

- **Evaluation Metrics and Tools**

  The paper proposes to use standard metrics and visualizations to evaluate the performance and precision of the models. In this case, the main metrics are [5]:

  1. Accuracy: Measures how correct your model predictions are by calculating the proportion of correct results among the total number of results.
  2. Precision: Measures the ratio of the number of correct predictions over all classes to the number of total predictions (Macro metric computes the average of the precision score of each separate class).
  3. Recall: Measures the ratio of the number of correct predictions over all classes when compared to the number of true samples (Macro metric computes the average of the precision score of each separate class).
  4. F1 score: Measures the harmonic mean of precision and recall (Macro metric computes the average of the F1 score of each separate class).

  In the same way, visualizations of model predictions are created using accuracy plot, violine plot, bar plot and heat map to illustrate the influence of different features on the model's performance.

4 Data Preprocessing

This section describes the dataset used and the pre-processing pipeline applied before training the models.

4.1 Dataset

FER-2013 [3] dataset is a common dataset used for facial expression recognition published by Kaggle. The dataset contains 35,887 gray-scale facial images categorized in 7 different emotions; *anger*, *disgust*, *fear*, *happiness*, *sadness*, *surprise* and *neutral*.

In figure 1, there is a subset of FER-2013 showing the first 10 for each emotion. This dataset contains non-human images and some random images, as you can see in the third image of the angry category or in the eighth image of the fear category in the figure 1.



Figure 1: Top 10 images per category in FER2013 dataset

4.2 Preprocessing

Since the FER-2013 [1] dataset is composed of a set of labeled folders with different images and contains many noisy and uninformative samples, data pre-processing and cleaning is performed first to transform the data into a format that is understandable to the models and to remove problematic observations that could lead the models to decrease their effectiveness.

4.2.1 Data Collection

The folder structure of the FER-2013 dataset divides the images first into train and test folders, which have the same internal structure, a set of folders of each category (*anger*, *disgust*, *fear*, *happiness*, *sadness*, *surprise*, *neutral*) where each folder contains the images labeled to that specific category.

To collect the data, the TensorFlow preprocessing package is used, which allows one to collect the data from different paths and automatically get the images with their target label, besides helping with the normalization and format of the images, resulting in a dataset object with all images with their respective target label in a format $48x48$ where each value is a pixel value normalized in a value between 0 and 1 (Divide the pixel value

by 255).

Finally, to have a more manipulable object, a pandas DataFrame is created with the image_path, image (Flattened array of pixel values), emotion_idx (Numerical value of each category), emotion_name, dataset (Train or test split) and the one hot encoding of the target that will be necessary for the model training task.

4.2.2 Data Distribution

As shown in figure 9. "Total number of image of each attributes with bar plot" in the paper [3], overall, the dataset is a little unbalanced, having a similar number of samples for categories "Angry", "Fear", "Neutral", and "Surprise", however, the classes "Disgust" and "Happy" bring out an imbalance having more than 8000 of samples for Happy, while Disgust samples are no more than 1000. This situation is considerable and can affect the learning phase of the different models, since they will tend to ignore this class since there is not enough evidence to extract features for this kind of samples.

Taking into account this aspect, and following the splitting described in the folder structure explained in the previous section, data are split across 3 different sets, train, validation, and test dataset (Equivalent to the division of Training, PrivateTest and PublicTest exposed in the paper). Validation and Test comes from the test folder in the original dataset, using the split function available in **sklearn**, with a proportion of 50% (Half of the test data) and stratification based on the sample target, a feature that ensures that both datasets remain a uniform distribution of samples per each category, avoiding datasets without categories pretty unbalanced (for instance, "Disgust").

4.2.3 Data cleaning

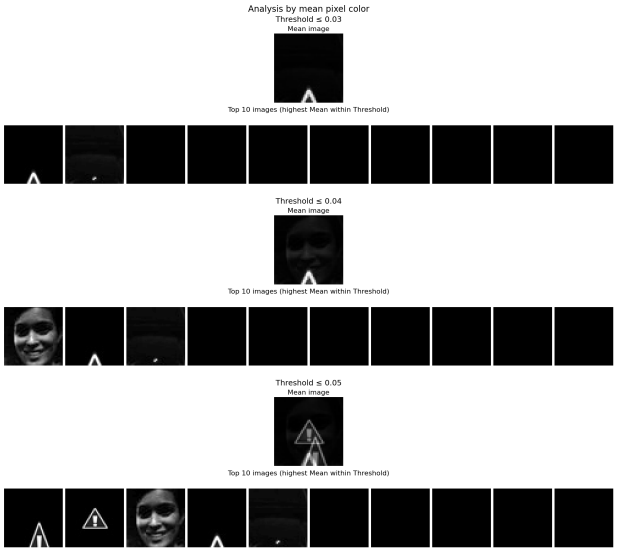
This section describes the dataset cleaning phase and exposes the resulted dataset that will be the data source for the models, as well as the effectivity of this cleaning step.

According to article [3], "Blank images" are the noise images found in the dataset. These images include any image where there is no recognizable human face, including text images, fully blank or white images, and improper images with very low quality. It is suggested that this removal task is performed through manual inspection or using automated methods such as image analysis algorithms or machine learning.

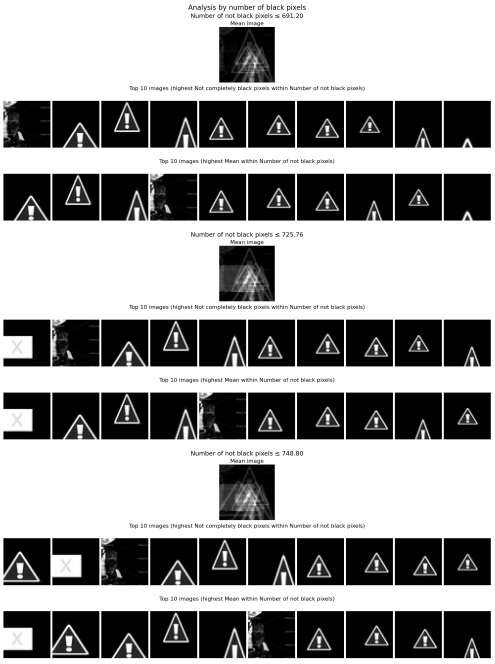
In this implementation, a strategy that combines manual inspection and automated methods is used, supported by a secondary function called *plot_analyze*, which based on the dataset, computes different metrics (defined by the user) for each image and filters them comparing with a threshold. The first approach was based on the darkness of the image, defined by the mean pixel color and the amount of "black" pixels ("Black" pixels based on the grayscale visualization) metrics, with which it is expected to remove any kind of black image without important information. Figure 2 displays the images discarded for each threshold with both metrics. In the image 2a, images with a mean pixel color lower

than 0.03 (8 in scale [0,255]) ensure that only black images are removed, but we also keep some images with an alert that do not mean anything to us in the recognition task. Therefore, as you can see in the image 2b, the amount of "black" pixels allows one to filter out improper images, using a threshold of 748 pixels, which means that any image with fewer than this number of pixels will be discarded.

Following the same idea, but focusing on the "whiteness" of the image, it is defined the mean pixel color in white (Mean of the resulting image from the difference between a totally black image and the face image) and the amount of "white" pixels ("White" pixels based on the grayscale visualization and the concept of "Black" pixel defined before) metrics. As a result, with both metrics in 3a and 3b, most text and icon images are removed, considering that their mean white pixel color is below 0.10 (25 in scale [0,255]) and their number of white pixels is lower than 690.

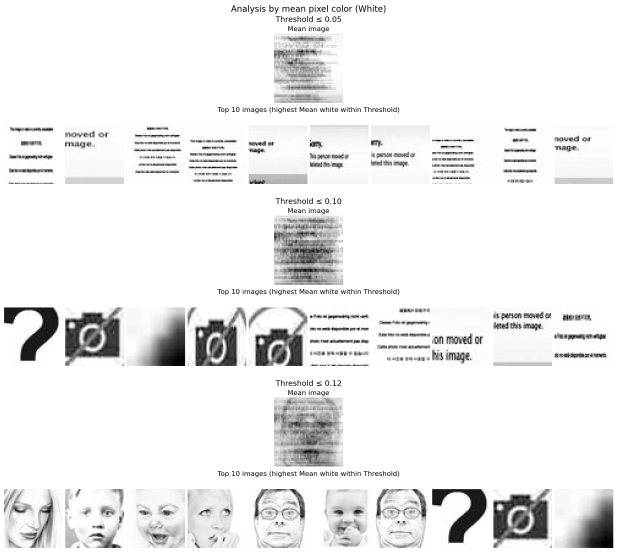


(a) Analysis by mean pixel color

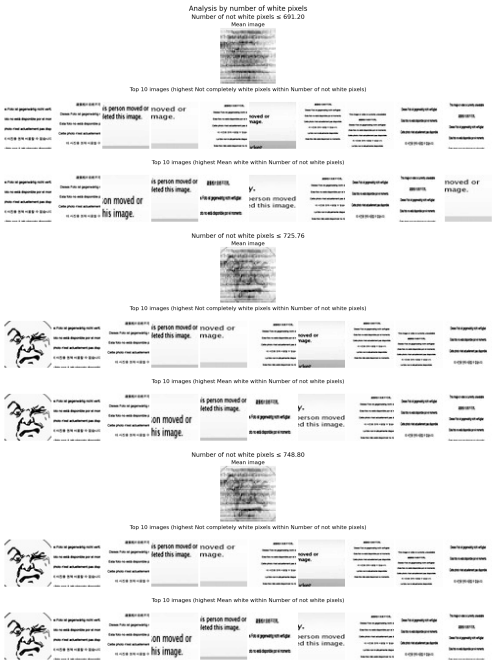


(b) Analysis by amount of "black" pixels

Figure 2: Image analysis based on their "blackness"



(a) Analysis by mean pixel color (White)



(b) Analysis by amount of "white" pixels

Figure 3: Image analysis based on their "whiteness"

Based on this strategy, the 53 images shown in figure 4 are removed from the original FER2013 dataset, resulting in a reliable dataset, useful in any training and validation task.

5 Models

This section describes the definition, implementation, and results for each of the models proposed in the paper.

5.1 Support Vector Machine

The first model is a Support Vector Machine (SVM), specifically, a Support vector classifier (SVC). The objective is to spot patterns and characteristics in the images, which allow us to find a set of hyperplanes that separate the seven classes with the largest possible margin.

First, Principal Component Analysis (PCA)is used to extract features from pictures of expressive faces. PCA was used to find the most important characteristics by reducing the dimensionality of the data. This phase is used to improve the efficacy of emotion recognition. The paper [3] proposes to use 67 fundamental components to feed into our support vector machine classifier, which according to the implementation (Figure 5), retains about 86.7% of the total squared singular-value mass (variance in the Frobenius sense). This



Figure 4: Blank images

indicates that the dominant structure of the data is largely captured by these components, while the remaining components contribute about 13.3% of the variance, indicating that this amount of dimension is enough to train the SVM and deduce the frontiers for each class.

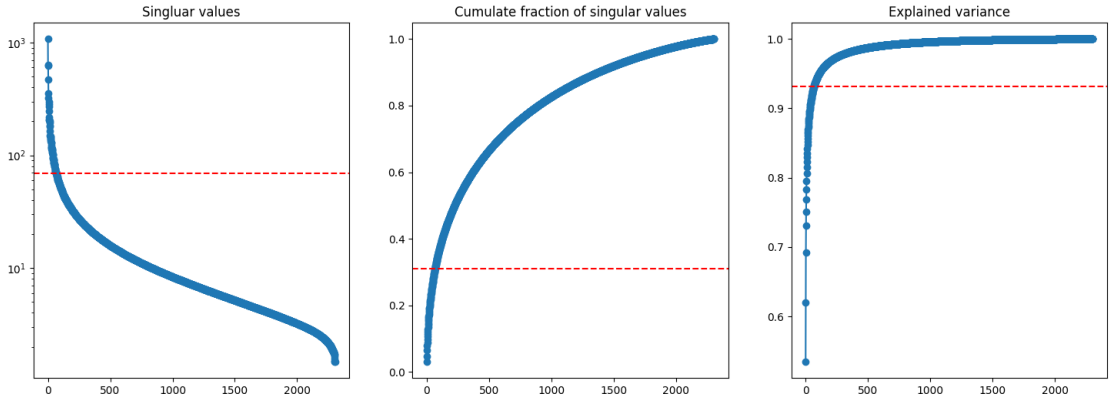


Figure 5: Singular Values of Principal Component Analysis

Given this number of Principal Components to consider, the original training, validation, and test data are projected onto this subspace, generating a new dataset with images of 67 features, which are later normalized and formatted to be appropriate to train and prediction tasks. The *sklearn* library is used to implement the class *SVC* which allows us to train and predict using a Support vector Classifier architecture; however, to improve and generate a good classification the phase of model tuning is essential to determine the optimal parameters which discriminate the images by different characteristics.

5.1.1 Model parameters

- **Kernel**:

  Depending on the kernel, the frontier or frontiers to classify the data changes. This process involves adding new features or transforming existing features to enrich and potentially add meaning to the data. The hyperplanes can then be calculated using the kernel function as if the dataset were represented in a higher-dimensional space. Based on the observations of the PCA of the dataset in the paper in Figures 16, 17 and 18, it is clear that we can ignore the 'linear' kernel, since the data do not look linearly separable. Consequently, performing an evaluation based on the F1-score, the kernel is selected among the following options: Polynomial kernel (Data transformed without additional dimensions to get polynomial boundaries), Gaussian kernel (It measures similarity between two data points in infinite dimensions and then approaches classification by majority vote), and sigmoid kernel (The decision boundary tries to separate the classes by fitting a sigmoid-shaped curve, resulting in a complex boundary that may not generalize well to unseen data).

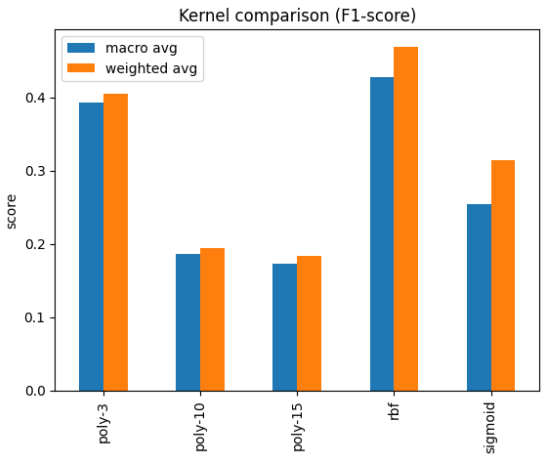
  After training on the data with the different kernel, and given that the F1 score provides a balance metric between accuracy and precision that reflects good learning and probably generalization, the selected kernel is the Gaussian kernel, providing a F1 score of 0.4 indicating moderate overall performance (Figure 7a).

- **Regularization Parameter**:

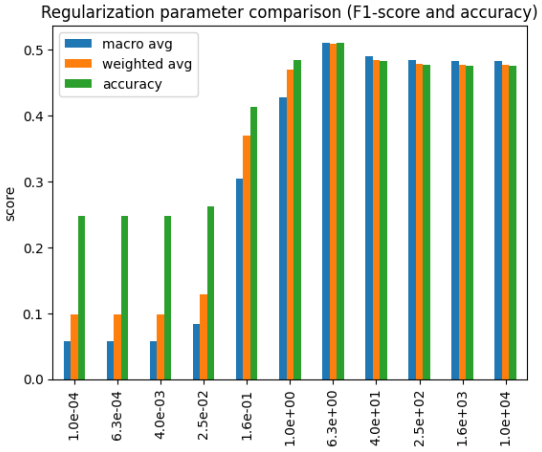
  Following the same strategy of the kernel parameter, by inspection, a regression parameter is selected given a definite range of values and knowing that in this case we implement an L2-penalty. As shown in Figure 7b, the regularization parameter that enables good learning is approximately 6.3, suggesting that a moderate regularization level is appropriate for our dataset.

- **Gamma Parameter**:

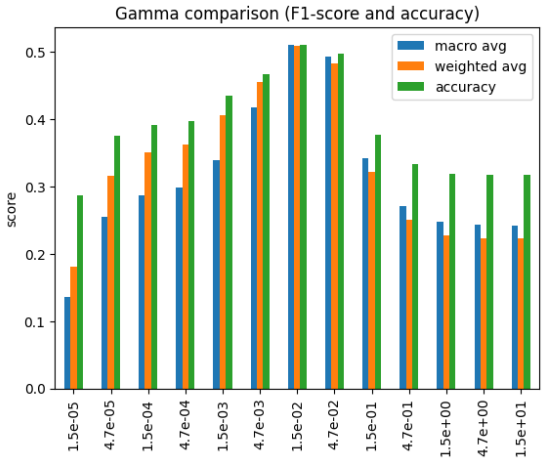
  Following the same strategies, by inspection and given the selection of the Gaussian kernel in the first parameter tuning, the suitable gamma parameter was selected, which controls the influence of each individual training sample on the decision boundary. As shown in Figure 6c, the gamma parameter that enables good learning in an unknown dataset (Validation dataset)is approximately 0.015, corresponding to smooth behavior to the boundaries, where the points need to be fairly close to be considered strongly similar.



(a) F1 score with different kernel



(b) F1 score with different values of regularization parameter



(c) F1 score with different values of gamma parameter

Figure 6: SMV parameters tuning

5.1.2 Results

Once the suitable parameters are selected, to ensure a good generalization and avoiding overfitting, one new model of Support Vector Classifier is trained and the following Confusion matrices describe the behavior of this simple model in the face recognition task.

Report Results on Train Dataset:

	precision	recall	f1-score	support
0	0.968136	0.915390	0.941025	3983.00000
1	0.995025	0.919540	0.955795	435.00000
2	0.967527	0.924731	0.945645	4092.00000
3	0.939907	0.985019	0.961934	7209.00000
4	0.926523	0.948512	0.937388	4972.00000
5	0.930915	0.946147	0.938469	4828.00000
6	0.983617	0.948799	0.965894	3164.00000
accuracy	0.948890	0.948890	0.948890	0.94889
macro avg	0.958807	0.941163	0.949450	28683.00000
weighted avg	0.949591	0.948890	0.948846	28683.00000

(a) Metrics on train dataset

Report Results on Validation Dataset:

	precision	recall	f1-score	support
0	0.368308	0.360587	0.364407	477.000000
1	0.781250	0.454545	0.574713	55.000000
2	0.404382	0.397260	0.400790	511.000000
3	0.598448	0.695603	0.643379	887.000000
4	0.476726	0.482927	0.479806	615.000000
5	0.430052	0.399679	0.414309	623.000000
6	0.765043	0.643373	0.698953	415.000000
accuracy	0.510745	0.510745	0.510745	0.510745
macro avg	0.546316	0.490568	0.510908	3583.000000
weighted avg	0.512061	0.510745	0.509119	3583.000000

(b) Metrics on validation dataset

Figure 7: Confusion matrix for support vector machine plot

These results make evident a generalization gap, showing that model is leaning patterns but they do not transfer well to the not public and private test dataset. However, an overfitting issue is almost discard given the fine tuning of each of the parameters and the robust and stable evaluation made in the data preprocessing and parameter tuning, resulting in that the better performance with a model of this kind is close to the value we get. In the same way, it is remarkable that some classes, such as 'Disgust' (1), 'Happy' (3) and 'Surprise' (6), are learned better than the other ones, supporting that is likely that more quality and amount of samples in some of the imbalanced categories can remove problems related with class imbalance. Overall, this model suggests that there is a feature limitation of PCA for FER2013, then it is viable to select a more complex approach to the classification task.

On the other hand, compared to the results exposed in the paper [3], it is clear that the prediction of this dataset is challenging, since they obtain higher precision values but consider a smaller test dataset, composed only of 337 samples. The paper states that feature extraction/processing can cause information loss and can reduce validation performance, supporting the results obtained in this implementation.

5.2 VGG19 Architecture

This second model is a Convolutional Neural Network inspired by a VGG19 model (Exposed in Figure 20 of the article [3]), with the objective of maximizing the accuracy of the model in the training data while minimizing damage. CNN's automatically detect the essential features without human supervision, making them stand out from other traditional classification algorithms. With all the extensive research in the field of CNNs in recent years, they have become a convenient tool to solve computer-vision tasks [6].

It is important to note that the implementation was made from scratch without using pretrained weights for the CNN. Therefore, to have real learning (Figure 9) and avoid information loss when we use ReLU, produced by the "Dying ReLU" problem exposed in the lectures, the initialization of the weights and the bias parameters use a ***He-normal*** initialization, except for softmax and the final layer. This decision is supported in article [7], which states that the combination of ReLU and He normal initialization outperforms

the other methods, has the lowest error rate most of the time and is specially effective in extremely deep networks, specifically in a biomedical image segmentation challenge named EM segmentation challenge, which supports the selection of this weight initializer for our CNN and our use case (Image face emotion classification).

5.2.1 Model tuning

- **Architecture in VGG19-style**:

  Based on the architecture exposed in the paper [3] and practically in the one displayed in Figures 20 and 21, a series of 5 blocks of convolutional and pooling layers is implemented, including a layer at the top where it transforms the data from grayscale to RGB, since the architecture expose and the ideal VGG19 has as input the RGB images and it is preferred to delegate this work to the neural network and his parameters and avoid it in the preprocessing stage. In the same manner, there is no definite specification of the architecture implemented in the paper after the convolutional blocks, therefore, based on different implementations, two different approaches are considered: (Figure 8a) use exactly a VGG19 architecture with a flatten layer and 2 dense layers with a final softmax layer, and (Figure 8b), which uses the convolutional part and adds just a flatten and predictions layer with softmax activation, expecting to avoid more complexity removing thousands of parameters and delegating more feature extraction to each convolutional layer. Finally, based on the description and suggestion exposed in the paper [7], a dropout layer is added between each of the fully connected layers at the end of the neural network, to prevent over-fitting and prioritizing the learning of reliable characteristics, not just fitting train data.

block5_pool (MaxPooling2D)	(None, 1, 1, 512)	0
flatten (Flatten)	(None, 512)	0
dropout_5 (Dropout)	(None, 512)	0
fc1 (Dense)	(None, 4096)	2,101,248
dropout_6 (Dropout)	(None, 4096)	0
fc2 (Dense)	(None, 4096)	16,781,312
dropout_7 (Dropout)	(None, 4096)	0
predictions (Dense)	(None, 7)	28,679

Total params: 38,935,623 (148.53 MB)
Trainable params: 38,935,623 (148.53 MB)
Non-trainable params: 0 (0.00 B)

(a) Exactly VGG19 architecture

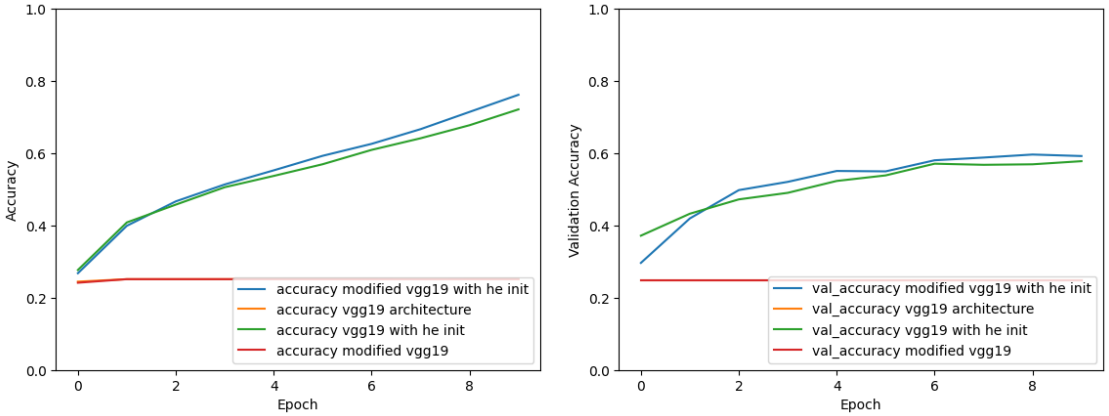
block5_pool (MaxPooling2D)	(None, 1, 1, 512)	0
flatten (Flatten)	(None, 512)	0
dropout_7 (Dropout)	(None, 512)	0
predictions (Dense)	(None, 7)	3,591

Total params: 20,027,975 (76.40 MB)
Trainable params: 20,027,975 (76.40 MB)
Non-trainable params: 0 (0.00 B)

(b) Modified version of VGG19 model

Figure 8: CNN architecture tuning

Given the parameters shown in the article and a short training period, it was concluded that the most efficient alternative to neural network architecture is version 2 (Figure 8b). Although the performance difference is not as big and almost insignificant, the behavior of the training using this architecture is stable, while a VGG19 from scratch tends to behave more erratic and changeable in the validation environment (Figure 9b). On the other hand, using a simplified model is preferred since the computing power to train a complete VGG19 is bigger, so an approximation could provide faster and quality results.



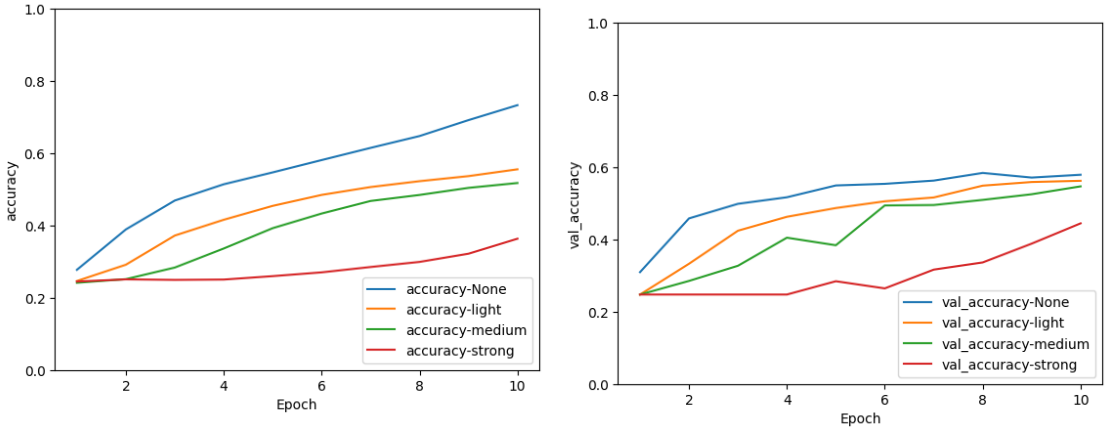
(a) Accuracy on train dataset (b) Accuracy on validation dataset

Figure 9: CNN architecture and initializer tuning

- **Augmentation**:

  As suggested by the paper, an augmentation layer is added at the top of the neural network. Those layers modify, to a greater or lesser degree, the images by applying different operations such as rotations and translations, with the objective of reducing overfitting and improving the robustness of the model, providing diversity in the data, forcing the neural network to recognize important features of the emotions and not only the faces in the training data.

  As shown in Figure 10, four different levels of augmenting were defined. Starting with a light augmenting level, which keeps the images close to the originals with small geometric changes. The medium level is based on moderate rotations, shifts, and zooms without heavily altering the content, while the high level performs an aggressive spatial augmentation (Rotation and Translation) plus contrast. Notice that it is also included a "None" level, to verify the natural performance of the neural network.



(a) Accuracy on train dataset

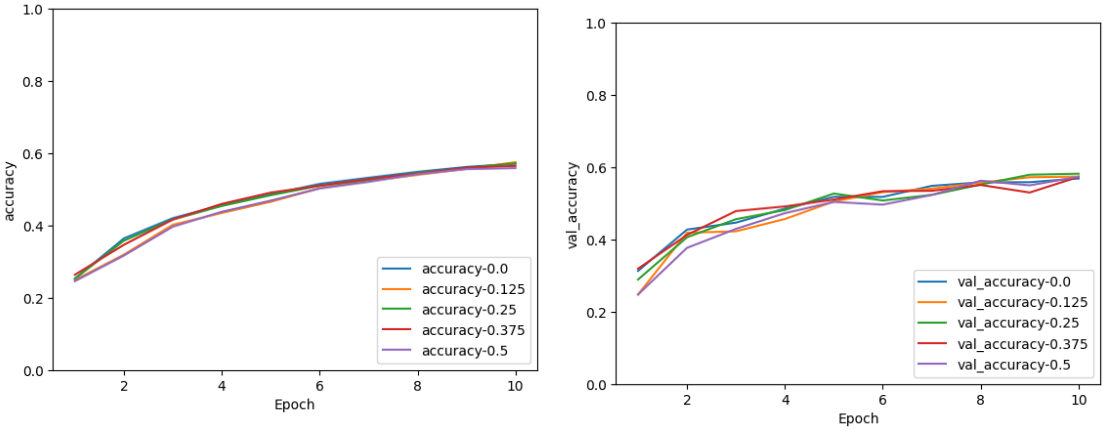
(b) Accuracy on validation dataset

Figure 10: Augmentation level tuning

The light augmentation level is selected, since as seen in figure 10b, this level provide a good and constant improvement on the unknown data (Training perspective), improving the generalization power of the model, while other augmentation levels, such as "None" looks like has an asymptomatic behavior, limiting the generalization power and allowing the presence of overfitting as shown in figure 10a. Then, a good balance between possible generalization and good performance over training data is given by the "Light" augmentation level.

- **Dropout rate**:

  Following the same idea of avoiding overfitting and improving the generalization power of the model, some drop out layers are used in the model, to avoid relying too heavily on specific neurons, encouraging it to learn more robust and generalized features. These layers should be tuned, by defining a suitable rate of neurons to deactivate during each forward and backward pass. Considering this statement about the function of the drop rate, the model is evaluated with different values in a range between 0 and 0.5, since it is consider to be excessive to have bigger values (More than a half of the neurons could be deactivated each pass).



(a) Accuracy on train dataset

(b) Accuracy on validation dataset

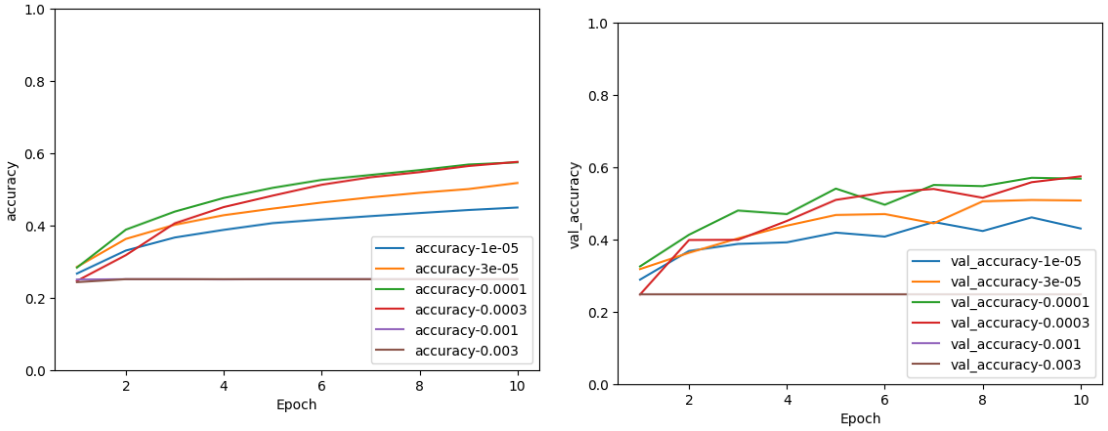
Figure 11: Dropout rate tuning

The results in Figure 11, shown that there is no significant difference with the selected drop rates, all of them achieving similar final values of accuracy and reporting similar behavior in the training as well as in the validation dataset. Therefore, the drop out layer are removed by simply choosing a 0.0 drop rate, taking advantage of the reduction on the model complexity that it is obtained.

- **Learning rate**:

  Following the same strategy used in the parameter tuning on the first model, by inspection, and given a range of possible values, a suitable learning rate was selected. The evaluation values were in a range between 1×10^{-5} and 1×10^{-3}, considering that the network is being trained from scratch and is a very complex network, then the update rule should be very stable and avoid abrupt changes.

  After training the proposed model with each of the learning rates in the range and evaluating its performance using validation accuracy, we selected a learning rate of 1×10^{-4}. This choice was made based on the validation accuracy, which serves as a balanced metric reflecting the model's ability to learn patterns effectively and generalize well, while minimizing the risk of overfitting to the training data. The chosen learning rate resulted in a final accuracy of approximately 57%, demonstrating stable performance without significant fluctuations or irregularities (Figure 12) compared to other values that keep changing or looks like the found a limit in their validation performance. This stability suggests that, with a larger number of epochs, the model would likely avoid overfitting and prevent erratic jumps near the minimum.



(a) Accuracy on train dataset

(b) Accuracy on validation dataset

Figure 12: Learning rate tuning

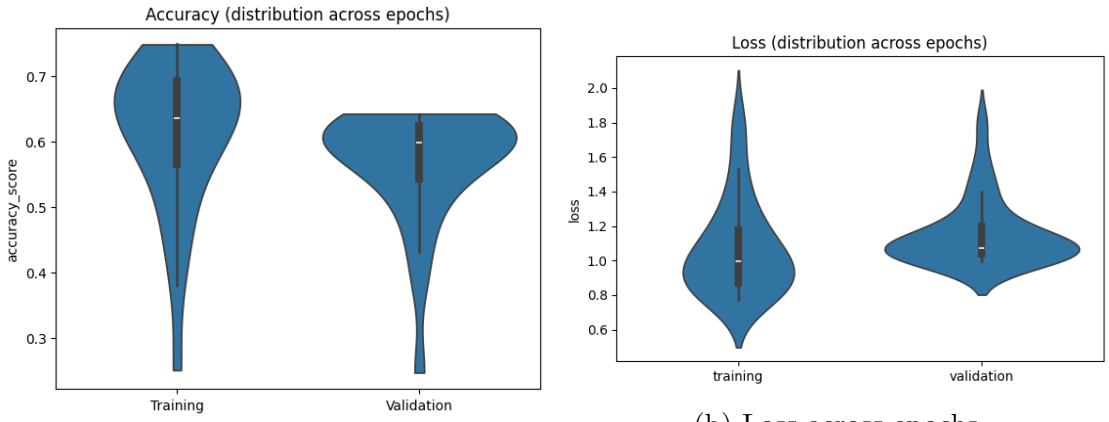
5.2.2 Results

Once the suitable parameters are selected, to ensure a good generalization and avoiding overfitting, one new instance of the CNN is trained with the parameters selected and the parameters suggested on the paper [3]. The results are exposed in Figure 13, Figure 14 and Figure 15.

To analyze the resulted accuracy and loss of the model after 30, first we focus on the behavior and distribution format of these 2 metrics. The violin plots in Figure 13 shows this distribution format behavior. Here the accuracy in the training dataset shows a median of approximately 0.65, with the inter quartile range lying between 0.56 and 0.69; while for the validation set, the median is almost 0.6, and the inter quartile range lies approximately between 0.54 and 0.62. This difference in the concentrations reveals a drop in the performance on unseen data. On the other hand, in the loss plot, the training set has a median of about 0.94, with an inter quartile range between 0.84 and 1.19, while the validation set has a median of approximately 1.09, with the inter quartile range lying between 1.03 and 1.14, revealing a very stable behavior on the validation data, having not so spread values in the training time, suggesting that there is no considerable overfitting.

This idea is also complemented with the trend of the accuracy and the loss over each epoch, shown in Figure 14, our accuracy score is monotonically increasing but our validation line is nearby of our training line, which shows an exponential growth until epoch number 10. Something similar and equivalent happens with the loss which decreases rapidly until epoch number 10, having very similar behavior in both datasets. After a given point near epoch 20, it is clear that the accuracy in training set keep trying to improve but in a very slow rate, while the validation performance stabilize (But fluctuates trying to grow), suggesting that overfitting could appear in later epochs. After 30 epochs, the model report an accuracy score of 0.64 and a f1 score of 0.64 in validation data, while 0.75 in both accuracy and F1 score for training data. These results suggest the presence of a slightly

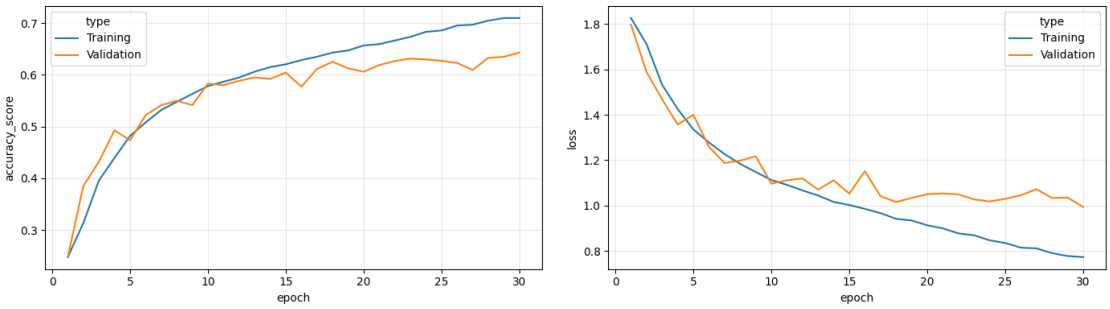
overfitting, but compared to results shown in the paper, the final performance is overall better, having a better generalization and a lower loss in the validation dataset.



(a) Accuracy across epochs

(b) Loss across epochs

Figure 13: Violine plot for accuracy And loss function.



(a) Accuracy across epochs

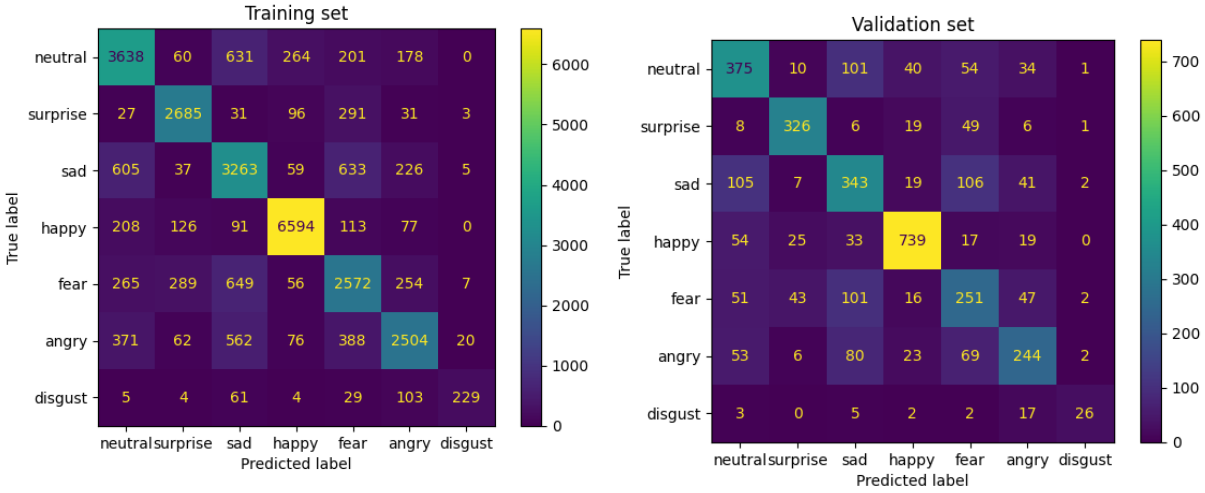
(b) Loss across epochs

Figure 14: Accuracy function plot and loss function plot.

Subsequently, in Figure 15 is visible the performance of the model over the different data sets, those confusion matrices break down the number of truth instances of a class against the number of predicted class instances. Firstly, consider all of them, it is a clear trend which states that the model performs really well to classify images corresponding to surprise, neutral, and happy faces. On the other hand, sad, angry, fear, and disgust images are harder to classify correctly, specifically for images related to disgust it is clear that the model isn't be able to learn anything because compared to the amount of data available of other emotions, disgust is pretty imbalanced (Mentioned before), then it is likely that most of the hits are just luck. In contrast, even though there were many samples of anger, sadness, and fear, there is great confusion between them, and analyzing the misclassified images (Figure 16), it is clear that it is not completely fault of the model, these emotions are fairly related and most of the images could be classified in any of those emotions, a clear consequence of the individual variations of facial expressions and the intrinsic ambiguity in the classification. Likewise, consider the misclassified images Figure 16, It is clear that the

model has problems in the identification for profile images, probably by it limited area to identify features characteristics of each expression.

Finally, just focusing on the miss predicted images in validation and train sets, it is clear that there exists a similarity and a common behavior, applying similar criterion for both classification, resulting with common miss classified images, revealing a clear generalization performed by the model.



(a) Metrics on train dataset

(b) Metrics on validation dataset

Figure 15: Confusion matrix for Model 2 with different datasets



(a) Miss-classified images in train set

(b) Miss-classified images in validation set

Figure 16: Top 10 miss-classified images

5.3 Convolutional network (Prove of hypothesis)

5.3.1 Objective

To address the conversation to the effectiveness and suitability of model 2 in the face expression recognition task, the article [7] suggests the use of "statistical testing". Statistical hypothesis testing is a statistical inference method that is used to decide whether the data provide sufficient evidence to reject or support a particular hypothesis. In this case, the performance of different models on the facial expression recognition task is used as evidence to support one of the following two hypotheses:

- **Null hypothesis:** There is no significant difference between the distribution of features extracted from the dataset (FER 2013) and the features extracted from a reference representation of general face images. This would suggest that CNN is mainly learning general facial structures rather than expression-specific features.
- **Alternative Hypothesis:** The features extracted by one CNN are able to capture unique expression-related features, and these features can be used to distinguish between different expressions with high accuracy. This would suggest that there is a difference between each the feature distributions associated with different face expressions.

To perform this comparison, the performance metrics used in the different model implementation are used as means of comparing both mean and covariance, following a t-test and ANOVA settings for the statistical test.

5.3.2 Implementation

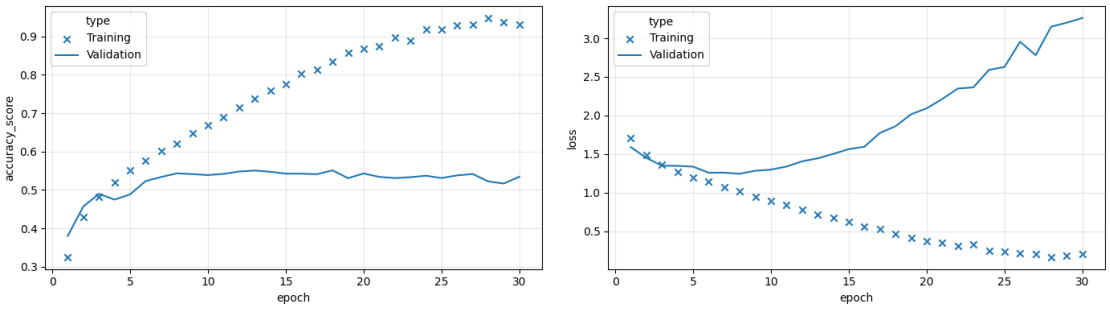
Considering the limitations in obtaining exact statistical indicators, we will focus on getting empirical results that support any of the hypotheses. Due to the fact that Model 2 explores the null hypothesis; Since it shows the performance of a VGG-style architecture in the learning of general expression patterns available in the data supplied by FER2013; The approach will focus on exploring the alternative hypothesis scenario with the implementation of two sequential convolution neural networks, a simple and a pretrained CNN.

The simple CNN model is based on a 3 block CNN (As shown in Figure 19 of the article [7]), and is intended to verify if the features extracted by a simple model, focusing on general details, can learn more meaningful features between the different facial Expressions. On the other hand, the pretrained CNN is used on a sizable dataset "ImageNet", using directly the TensorFlow implementation of VGG19 with the pretrained weights on that dataset. The network is then fine-tuned on the FER2013 dataset and, using the pretrained features (On a general reference dataset) and specializing it in the facial recognition task (Fine tune stage), the model will reflect if a more powerful and complete model is able to learn the distribution features that can differentiate effectively facial expressions.

As a result, if any of those models results in better performance on unseen data, compared to model 2 (Which explores null hypothesis), the alternative hypothesis would be empirically supported, and the null hypothesis could be discarded, following the idea that more complex and powerful CNN could get better performance.

Both approaches are implemented following the same methodology described in the previous models, the model architecture is defined, a hyperparameter tuning stage is used and the model is evaluated against a validation set.

For the first approach (The simple model), in hyperparameter tuning, a learning rate of 10^{-3} was selected to the training stage, with the same number of epochs and optimizer used in the implementation of model 2. The results keep the trend of the ones shown in the source article, having a clear improvement in performance (Based on the accuracy and the F1 score) in training dataset, but getting an asymptotic limit on performance for the validation dataset (Figure 17). Moreover, it is clear that the model becomes worse each epoch, with a trend of increase in the loss of validation data, while training loss keeps decreasing, indicating that the model is not able to learn patterns in the data to improve its accuracy.



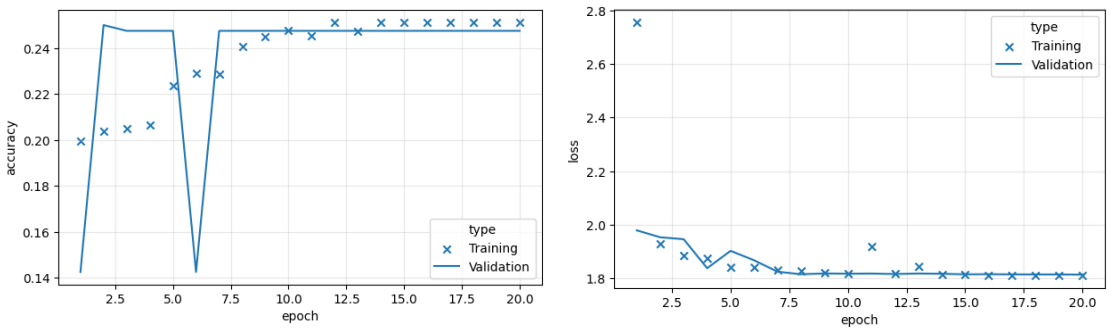
(a) Accuracy Function Plot (b) Loss function plot

Figure 17: Behavior of model 3 during train time



Figure 18: Prediction distribution

On the other hand, for the second approach, the hyperparameter tuning yielded a value of 4×10^{-4} for the learning rate parameter. Likewise, for this implementation pipeline, the training stage is divided in an initial training; to set values to the top layers added manually to the pretrained network (First 10 epochs); and a fine tuning part, where the pretrained layers are allowed to be trained following the target values on the FER2013 dataset. As a result of the learning stage of the model, Figure 19 shows that the model cannot use the previous data to generalize and identify facial expressions with FER2013 data, reaching a constant performance (24,76%) after epoch 8. In like manner, considering the inference and prediction work performed by the model (Figure 20), it is notable that the distribution of the results (Probabilities represented in the final softmax layer) is similar for all images, suggesting a unique class prediction, exposing zero learning.



(a) Accuracy Function Plot

(b) Loss function plot

Figure 19: Behavior of model 3 pre-trained during fine tune time



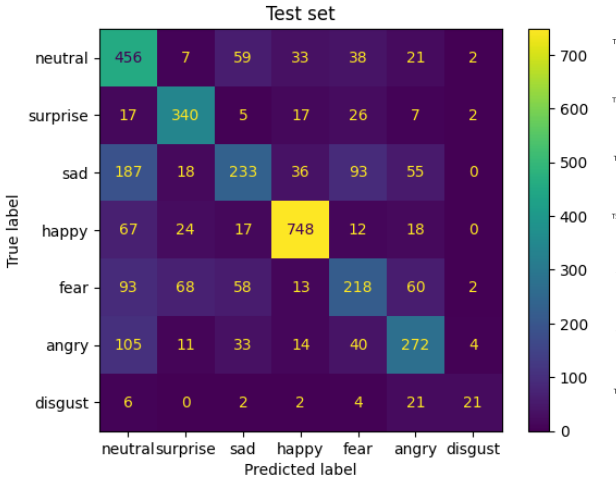
Figure 20: Prediction distribution

5.3.3 Results

Consider the resulting performance of the models and the similarity of the decomposition of each prediction (Bar graphs of the predicted images); there is no empirical support for the alternative hypothesis. Instead, they suggest that the observed improvement is limited to in-sample fitting and does not translate into reliable performance on unseen data, supporting the idea that there is no significant difference between the feature distributions of the dataset and the reference face images (Null hypothesis). Consequently, based on the statistical indicators selected and the experimentation, the alternative hypothesis is not supported, and model 2 should be preferred because of its power of generalization. Likewise, following the idea of the paper, if you intend to improve the prediction accuracy, you would have to concentrate more on data pre-processing instead of convolutional neural network.

6 Results Comparison

After comparing models, using the statistical power of cross-validation (Test, validation and training subsets) and based on the empirical hypothesis testing performed with model 3, the final model 2 was trained for 30 epochs with a learning rate of 3×10^{-4} and the complete data set (Validation + training). It shows a training accuracy of 71,8% and a testing accuracy of 63,39%, following the trend mentioned in the Model 2 analysis. Similarly, the generalization is notable when the confusion matrix (Fig 21a) is observed, where it is verified that a better relative classification is done for the happiness and neutral expression, and exposing problems with the differentiation of angry, sadness, and fear expressions (Specifically differentiate neutral and sad images, as well as sad and fear images), where some images could be misinterpreted by different persons with different criteria (Figure 21b); therefore, these mis-predictions are not associated with the learning process; instead, it is related with a human intrinsic issue.



(a) Confusion matrix



(b) Top 10 images miss-predictions

Figure 21: Final model performance (Compared to test images)

Comparing these results with the final performance of the model exposed in the particle [3], they are notably similar, the final generalization accuracy is close to 64%. However, the model and training stage implemented across this project is more robust and complete, describing more strategies used to avoid overfitting, as well as using all the datasets for validation and testing; This robustness is notable in the final confusion matrix, where the diagonal (Correct predictions) have a larger impact on the overall performance, compared to the final confusion matrix (Figure 27 in the article), where it has powerful recognition on the happiness, sadness, surprise, and neutral categories, but the full matrix displays more miss-predictions, revealing a not balanced recognition, focusing on predict more frequently the expression with the largest amount of samples in the dataset.

On the other hand, it is important to compare the results of each implemented model and the correct interpretation of their results with the complete recognition task.

- Model 1

  Being the simplest and smaller, it provides a good benchmark and an initial point from which the performance should improve. Although the results exposed in [3] are calculated on a different dataset (Look at the support column), the training phases are similar, getting a high performance in the classification with the first 67 Principal Components. However, in our implementation, it is clear that the generalization power of the model is not so reliable, getting just a 51% of the f1-score (Balanced metric) in the validation set, a very decent performance for the task but not reliable for a real prediction environment.

  In this section, it is important to remark also the justification and validation of the selection for the input dimension. In paper [3], the 67 First Principal Components are stated as the right selection for it usability; However, with the Principal Component Analysis implemented, it was clear that this value was a good trade-off between the information lost and the simplicity for a simple model, such as Support Vector Machine.

- Model 3

  Compared to the base article [3], the model 3 and hypothesis testing was not well explained and the use of the model was not clear, although their results were appropriate to draw some conclusions. With the implementation, the objective of these models was clearer, to validate the idea that model 2 was the most suitable, and most importantly, the results following the same trend, having a simpler model that depicts a clear overfitting and not supports the idea stated by the alternative hypothesis.

7 Limitations

During the study of the facial image expression recognition task, there will be 3 main limitations that we met, the facial expression recognition problem, dataset issues, and limitations related to computational power.

- **Facial expression recognition problem**

  In addition to quality, illumination, pose, and angle of view, the main problem for facial expression recognition is related to subject-specific differences. Professor Elena Walsh [8] states that there are three problems when you think person's inner emotions can be accurately and reliability predicted based on facial expressions. First, emotions do not always produce the corresponding facial expression. Sometimes, people use expressions to conceal true feelings or pretend to possess them. Second, emotions are difficult to define, and the prototypical expression associated with an emotion can be infrequent or completely absent in 'real world' situations. And third, emotions and expressions can vary significantly across contexts, including cultural contexts.

  Clearly, these issues are completely intrinsic to humans, thus it is just an idealization thinking about a machine recognizing human expressions by an image, without knowing the context and without knowing the characteristics of each human. Then, this natural problem brings out a higher level of complexity to the problem, providing a new perspective of the results of the models implemented.

- **Dataset**

  Throughout the project, one of the biggest limitations was the quality of the FER dataset. Although it provides variety and good quality in the images, as well as different instances of the same expressions, there was a clear imbalance on the classes, generating an imbalanced training and therefore biased predictions. Similarly, the classification of the images was inadequate, providing a lot of miss predictions in images that could be perfectly classified in another class. However, this problem is closely related with the first limitation, but considering the use of the data, a better selection of the images (Following standards) could have been done to facilitate the recognition task.

- **Computational power**

  The most important and probably the one with the biggest impact limitation was the computational power. When you work with these complex models, with millions of parameters, the training and model tuning stage becomes reasonably demanding. One of the main limitations occurs in the tuning model phase, since it was not possible to perform a complete cross-validation of the parameters, resulting in a tuning by stages where each hyperparameter is tuned one at a time.

  Although this simplification allows us to achieve good results in a reasonable period of time, this could bring with it a reduction in the performance of the optimal model that could have been found with complete cross-validation. However, the train-validation-test split provides a powerful statistic measurement that allows a well supported selection of the best model possible and the fair comparison between them

8 Conclusion

The implementation of the work related to facial expression recognition task, conducts to different conclusions in different aspects:

1. The implementation of multiple models and the train-validation-test split provide a robust baseline to infer general behavior and performance for different models, and it provides us with a statistical rigor that allows us to discard and support different hypotheses made over multiple models.

2. Using a VGG19 like Convolutional Neural Network trained in our training and validation test, a performance of approximately 63% (F1 Score of 63,18%) is shown in the test data (Completely unseen dataset) after 30 epochs, being able to classify almost all face expression images associated with surprise and happiness. However, a difficulty in recognition of sad, fear, and disgust expressions being persistently evident, mostly related to the quality of FER2013 dataset and the emotional ambiguity described through the project.

3. With the statistical hypothesis testing performed, new models were trained to demonstrate their inference power and the distribution of individual prediction scores influenced by the different expressions worked on the project. These results do not provide any reason to support the alternative hypothesis and consequently reject the null hypothesis, which provides a statistical source to validate the effectiveness of model 2 in the expression recognition task and suggests that the optimal way to improve the prediction accuracy is preprocessing and image quality selection in the FER2013 dataset.

9 References

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